**The Solution:**

The proof breaks down to two statements.

1. If $\beta$ is a realized number, then $\beta > \sqrt{3}$.
2. For any $\epsilon > 0$ there exists a realized number $\beta < \sqrt{3} + \epsilon$.

# 1 Proof of Statement 1

**Overview:** We set $G = G_\beta$. The argument is very much like in my paper, *The Optimal Paper Moebius Band* (Annals of Math 2025). The connection to Moebius bands is that the quotient $\Sigma/G$ is a flat Moebius band, and $f$ induces a map from $\Sigma/G$ to $\boldsymbol{R}^3$. The main idea is to find two "crosscutting" line segments $(A, B)$ in the strip $\Sigma$, having their endpoints in $\partial\Sigma$, such that $f(A)$ and $f(B)$ are perpendicular and coplanar line segments with disjoint interiors that are respectively longer than $A$ and $B$. We then compare the trapezoid bounded by $A$ and $G(A)$, a fundamental domain for the action of $G$, with the geometry forced by $f(A)$ and $f(B)$.

**Crosscuts:** The $G$-*orbit* of a subset or a point of the strip $\Sigma = \boldsymbol{R} \times [0, 1]$ is the image of that subset or point under the action of $G$. Our second condition on $f$ is that $f(p) = f(q)$ if and only if the points $p$ and $q$ are in the same $G$-orbit. We call our clean triangulation $\mathcal{C}$ and we call the triangles of the clean triangulation $\mathcal{C}$-*triangles*. Because $\mathcal{C}$ is locally finite, the $\mathcal{C}$-triangles are ordered left to right, with consecutive triangles sharing an edge.

Recall that the *distinguished edge* of a $\mathcal{C}$-triangle is the edge in $\partial\Sigma$. The *distinguished vertex* of a $\mathcal{C}$-triangle is the vertex opposite the distinguished edge. This is the vertex in the other boundary component. Say that a *crosscut* is a line segment $A \subset \Sigma$ that joins the distinguished vertex of a $\mathcal{C}$-triangle $\tau$ to a point on the distinguished edge of $\tau$. The interior of the strip $\Sigma$ is foliated by the (relative) interiors of the crosscuts.

Given two crosscuts $A$ and $B$ we write $A \prec B$ is the midpoint of $A$ lies to the left of the midpoint of $B$. (Both these midpoints lie on the centerline of $\Sigma$.) We call the pair $(A, B)$ *admissible* if $A \prec B \prec G(A)$. This condition implies that $A$ and $B$ have different $G$-orbits. Note that $(A, B)$ is admissible if and only if

$$I(A, B) := (B, G(A)) \tag{1}$$

is admissible. The map $I$ is a self-map of the set of admissible pairs of crosscuts.

**Lemma 1.1** *If $(A, B)$ is an admissible pair, then $f(A)$ and $f(B)$ have disjoint interiors.*

**Proof:** If not, then there are interior points $p \in A$ and $q \in B$ such that $f(p) = f(q)$. But then the $G$-orbit of interior($A$) intersects the $G$-orbit of interior($B$). Since our crosscuts have disjoint interiors, this means that $A$ and $B$ have the same $G$-orbit. Contradiction. ♠

**Lemma 1.2** *$f$ increases arc length on each crosscut.*

**Proof:** Recall that $f$ is a squeeze map on each triangle $\tau$. So $\phi := f|_\tau$ is affine, and $\phi$ decreases the length of the distingished edge of $\tau$ and increases the lengths of the other edges. Let $p$ be the distinguished vertex of $\tau$ and let $q$ be any point in the distinguished edge. We just need to see that $\phi$ increases the distance from $p$ to $q$.

Without loss of generality we can translate the picture so that $p$ and $f(p)$ are the origin respectively in $\boldsymbol{R}^2$ and in $\boldsymbol{R}^3$. Then $\phi$ is a linear transformation. Setting $V' = f(V)$, etc, this lemma is equivalent to the following. We have vectors $V, W, V', W'$ such that

$$\|V\| < \|V'\|, \qquad \|W\| < \|W'\|, \qquad \|V - W\| > \|V' - W'\|, \tag{2}$$

and we want to see that $\|(1-t)V + tW\| < \|(1-t)V' + tW'\|$ for all $t \in (0,1)$. This last inequality is true provided that $V \cdot W < V' \cdot W'$. Squaring the inequalities in Equation 2,

$$2V \cdot W = \|V\|^2 + \|W\|^2 - \|V - W\|^2 < \|V'\|^2 + \|W'\|^2 - \|V' - W'\|^2 = 2V' \cdot W'.$$

So, indeed, $V \cdot W < V' \cdot W'$. ♠

**The Key Topological Idea:** This proof has the same ideas as Lemma T in my paper.

**Lemma 1.3 (T)** *$\Sigma$ has a pair $(A, B)$ of admissible crosscuts such that $f(A)$ and $f(B)$ are perpendicular and coplanar.*

**Proof:** Let $\mathcal{A}$ denote the set of all admissible pairs. We parametrize the set of crosscuts by $\boldsymbol{R}$: Each crosscut is parametrized by the $X$-coordinate of its midpoint. With this parametrization, $\mathcal{A} \subset \boldsymbol{R}^2$ is the interior of the closed infinite strip $\overline{\mathcal{A}}$ bounded by the lines $y = x$ and $y = x + \beta$. We call these lines respectively $\partial_0\mathcal{A}$ and $\partial_1\mathcal{A}$. The map $I$ from Equation 1 acts as a glide reflecton on $\mathcal{A}$, namely $I(x, y) = (y, x + \beta)$.

For any subset $X \subset \Sigma$ let $X' = G(X)$. We direct each crosscut $C$ so that it points upwards. We orient $f(C)$ so that $f$ is direction preserving on $C$. Let $\overrightarrow{f}(C)$ be the vector that points in the direction of $f(C)$ and has the same length. Note that $\overrightarrow{f}(C') = -\overrightarrow{f}(C)$.

Let $(A, B)$ be a pair of crosscuts, with midpoints $a$ and $b$ respectively. Using the dot product and cross product, define

$$g(A, B) = \overrightarrow{f}(A) \cdot \overrightarrow{f}(B), \qquad h(A, B) = (f(a) - f(b)) \cdot (\overrightarrow{f}(A) \times \overrightarrow{f}(B)). \tag{3}$$

By construction, $g$ and $h$ are continuous functions on $\overline{\mathcal{A}}$. Furthermore, if $g(A, B) = 0$ then $A$ and $B$ are perpendicular. If $h(A, B) = 0$ then $A$ and $B$ are coplanar. (If $f(A)$ and $f(B)$ are both parallel to the $XY$-plane then $h(A, B)$ measures the difference in the $Z$-coordinates of these segments.) In view of this, it suffices to prove that $(0, 0) \in \Phi(\mathcal{A})$, where $\Phi = (g, h)$.

Using $f(a') = f(a)$ and $\overrightarrow{f}(C') = -\overrightarrow{f}(C)$ and the symmetries of the dot and cross products, we establish the *oddness property*: $\Phi \circ I = -\Phi$. Here is the derivation.

$$g(B, A') = \overrightarrow{f}(B') \cdot \overrightarrow{f}(A) = -\overrightarrow{f}(B) \cdot \overrightarrow{f}(A) = -g(A, B).$$

$$h(B, A') = (f(b) - f(a')) \cdot (\overrightarrow{f}(B') \times \overrightarrow{f}(A)) = (f(b) - f(a)) \cdot (\overrightarrow{f}(A) \times \overrightarrow{f}(B)) = -h(A, B).$$

Note that $h = 0$ on $\partial\mathcal{A}$. Because $\overrightarrow{A}$ and $\overrightarrow{B}$ are parallel on $\partial_0\mathcal{A}$, we have $g > 0$ on $\partial_0\mathcal{A}$. Likewise $g < 0$ on $\partial_1\mathcal{A}$. Hence $\Phi(\partial_0\mathcal{A}) \subset X_+$, the positive $X$-axis, $X_+$. Likewise $\Phi(\partial_1\mathcal{A}) \subset X_-$, the negative $X$-axis.

Suppose $(0,0) \notin \Phi(\mathcal{A})$. If $\gamma \subset \overline{\mathcal{A}}$ is a segment with endpoints in the two boundary components of $\mathcal{A}$ then $\Phi(\gamma)$ is a continuous arc that joins a point in $X_+$ to a point in $X_-$ and misses the origin. Therefore, $\Phi(\gamma)$ has a well-defined half-integer winding number $w(\gamma)$ with respect to the origin. If $\gamma$ is such a segment, then so is $\gamma^* = I(\gamma)$. On the one hand, $\gamma^*$ and $\gamma$ are isotopic relative to $\partial_0\mathcal{A}$ and $\partial_1\mathcal{A}$ and $w(\cdot)$ is continuous and half-integer valued. Hence $w(\gamma) = w(\gamma^*)$. On the other hand $\Phi(\gamma^*) = -\Phi(\gamma)$, by the oddness property. But then $w(\gamma^*) = -w(\gamma)$, a contradiction. Hence $(0,0) \in \Phi(\mathcal{A})$. ♠

**The Trapezoid:** Let $(A, B)$ be as in Lemma T. The images $f(A)$ and $f(B)$ are perpendicular and coplanar and by Lemma 1.1 have disjoint interiors. Either the line extending $f(A)$ is disjoint from the interior of $f(B)$ or the line extending $f(B)$ is disjoint from the interior of $f(A)$. If necessary we replace $(A, B)$ by $I(A, B)$, which also satisfies Lemma T, to arrange that the line extending $f(A)$ is disjoint from the interior of $f(B)$. Pre-composing and post-composing $f$ by isometries we can arrange, as in Figure 1, that

- $f(A)$ lies in the $X$-axis and $f(B)$ is contained in the non-positive portion of the $Y$-axis.
- $f$ maps the upper vertex of $A$ to the left vertex of $f(A)$.
- $f$ maps the top vertex of $B$ to the top vertex of $f(B)$.

Let $t \in \boldsymbol{R}$ be such that $1/t$ is the slope of $A$. Figure 1 depicts the case when $t > 0$, and $f(A) \cap f(B) = \emptyset$, but our argument does not need these conditions.

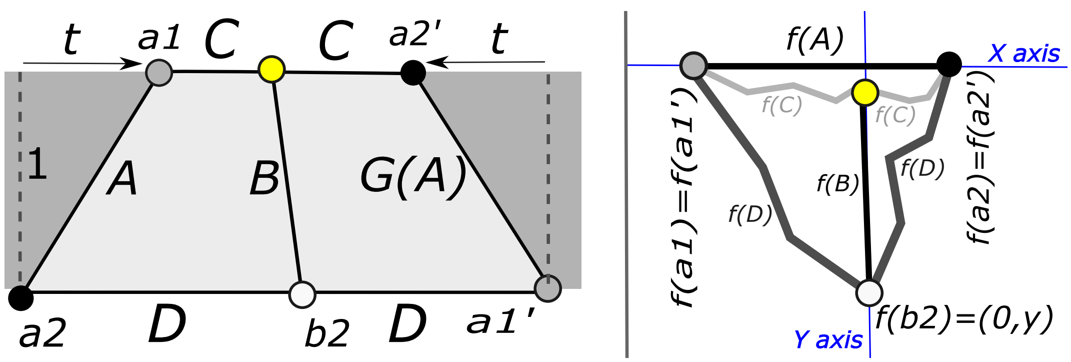


**Figure 1:** The domain is on the left and the range is on the right.

Let $C$ and $D$ be the segments in $\boldsymbol{R} \times \{1\}$ and $\boldsymbol{R} \times \{0\}$, respectively, connecting the endpoints of $A$ and $G(A)$. Since $G$ is a glide reflection, the trapezoid $T$ bounded by $A, C, g(A), D$ is isosceles. Starting from the bottom left and going clockwise, the vertices of $T$ are $a_2, a_1, a_2', a_1'$. We have $G(a_k) = a_k'$ for $k = 1, 2$. In other words, $G$ identifies the vertices in opposite corners of $T$. Hence, so does $f$. Also, $b_2$ denotes the bottom endpoint of the bend $B$. (The top endpoint of $B$ is colored yellow in the figure but has no name.)

Let $\ell$ denote arc-length. Note that $\ell(A) = \sqrt{1+t^2}$ and $\ell(B) \geq 1$.

**Lemma 1.4** *$\ell(C) > \sqrt{1+t^2}$ and $\ell(D) > \sqrt{5+t^2}$.*

**Proof:** By Lemma 1.2, the map $f$ increases arc length on each crosscut. The squeeze map property implies that $f$ decreases arc length on $\partial\Sigma$. Hence

$$\ell(f(A)) > \ell(A) = \sqrt{1+t^2}, \quad \ell(f(B)) > \ell(B) \geq 1, \quad \ell(C) > \ell(f(C)), \quad \ell(D) > \ell(f(D)).$$

Since $f(C)$ connects the endpoints of $f(A)$, we have $\ell(f(C)) \geq \ell(f(A))$. All together,

$$\ell(C) > \ell(f(C)) \geq \ell(f(A)) > \ell(A) = \sqrt{1+t^2}.$$

The point $f(b_2) = (0, y)$ separates $f(D)$ into two arcs $\gamma_1$ and $\gamma_2$. Let $\gamma_2^*$ denote the reflection of $\gamma_2$ in the horizontal line through $(0, y)$. We have

$$\ell(D) > \ell(f(D)) = \ell(\gamma_1 \cup \gamma_2) = \ell(\gamma_1 \cup \gamma_2^*) \geq \sqrt{4y^2 + \ell(f(A))^2} \geq \sqrt{5+t^2}.$$

The second equality comes from symmetry. The final inequalities come from the fact that $\gamma_1 \cup \gamma_2^*$ connects two points which are separated horizontally by $\ell(f(A)) > \sqrt{1+t^2}$ and vertically by $|2y| \geq 2\ell(f(B)) > 2$. ♠

**Endgame:** Given our definition of $t$, Lemma 1.4 gives

$$\beta = \ell(C) + t > u(t) := \sqrt{1+t^2} + t,$$

$$\beta = \ell(D) - t > v(t) := \sqrt{5+t^2} - t. \tag{4}$$

Hence $\beta > \max(u(t), v(t))$. Let $t_0 = 1/\sqrt{3}$. Note that $u(t_0) = v(t_0) = \sqrt{3}$, and $u$ is an increasing function and $v$ is a decreasing function. Hence $\max(u(t), v(t)) \geq \sqrt{3}$ for all $t$. Hence $\beta > \sqrt{3}$. This proves Statement 1.

# 2 Proof of Statement 2

**Overview:** Let $\epsilon > 0$ be given. We will produce a realized number $\beta < \sqrt{3} + \epsilon$. We take $\epsilon < 1/100$ for convenience. We build a polyhedral Moebius band $\Omega$ by stacking three equilateral triangles on top of each other and connecting them in a cyclic pattern using very thin rectangles. We then extract the combinatorial details and construct our map $f : \Sigma \to \boldsymbol{R}^3$ having $\Omega$ as the image. So, we first describe the range, then the domain, then the map.

**The Range:** Let $\mu = 2/\sqrt{3}$. This is the side length of a clean equilateral triangle. Let $T_0 \subset \boldsymbol{R} \times (-\infty, 0] \times \{0\}$ be the equilateral triangle, of side length $\mu + \epsilon^2$, which has an edge centered at the origin and contained in the $X$-axis. Let $\partial_- T_0$, $\partial_+ T_0$ and $\partial_0 T_0$ respectively denote the left, right, and top boundary edge of $T_0$. See Figure 2 below.

Let $E_0$ denote the segment $\partial_0 T_0 + (0, \epsilon^3, 0)$. We are translating $\partial_0 T_0$ parallel to itself and slightly off itself. Figure 2 (left) shows $T_0$ and $E_0$ fairly accurately. Here we are looking down on $\boldsymbol{R}^2 \times \{0\}$.

Define $T_\pm = T_0 \pm (0, 0, \epsilon^3)$. These two triangles lie in horizontal planes just above and below $T_0$. We label the edges of $T_\pm$ so that $\partial_j T_\pm$ is parallel to $\partial_j T_0$. Here $j \in \{0, +, -\}$.

We now define 3 sets, $R_+$, $R_-$, and $R_0$. We call these *connectors*.

- $R_\pm$ is a thin rectangle, the convex hull of $\partial_\pm T_0 \cup \partial_\pm T_\pm$. The long sides of $R_\pm$ have length $\mu + \epsilon^2$ and the short sides have length $\epsilon^3$.

- $R_0 = S_+ \cup S_-$ is the union of two rectangles. Here $S_\pm$ is the convex hull of $\partial_0 T_\pm \cup E_0$. The long sides of $S_\pm$ have length $\mu + \epsilon^2$ and the short sides have length less than $2\epsilon^3$.

Note that $R_\pm$ connects $\partial_\pm T_0$ to $\partial_\pm T_\pm$. Likewise, $R_0$ connects $\partial_0 T_-$ to $\partial_0 T_+$ while remaining disjoint from $T_0$. We used $E_0$ to slightly bend $R_0$ away from $T_0$. Define

$$\Omega = S_- \cup T_- \cup R_- \cup T_0 \cup R_+ \cup T_+ \cup S_+. \tag{5}$$

Figure 2 (right) shows this union schematically.

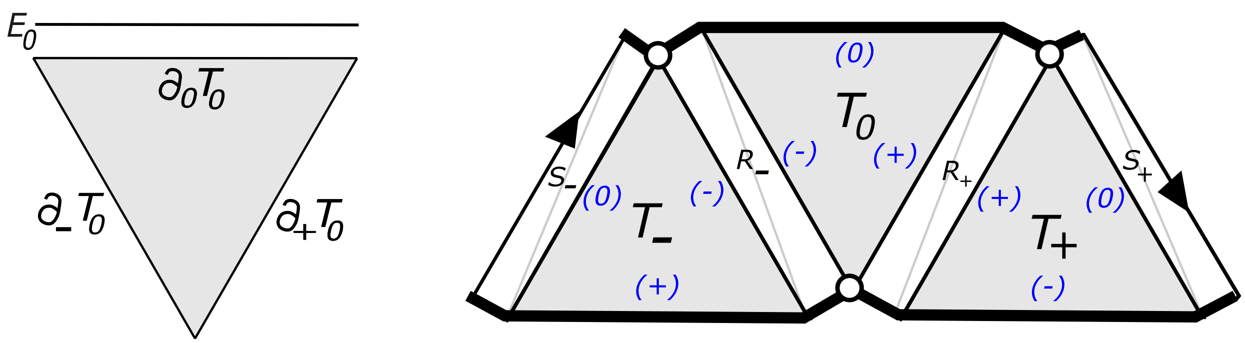


**Figure 2:** Constructing the range.

With the pieces ordered cyclically according to the listing in Equation 5, consecutive pieces intersect precisely along an edge of length $\mu + \epsilon^2$. The remaining pairs of pieces are disjoint. Hence $\Omega$ is a piecewise affine surface with boundary. Figure 2 (right) shows this, with the edge labels abbreviated. To get from Figure 2 (right) to the true picture in $\boldsymbol{R}^3$, we fold $T_-$ under $T_0$ and we fold $T_+$ over $T_0$. With the gluing diagram shown in Figure 2 (right), this folding makes the arrows line up in the same direction along $E_0$, as they should. So, the gluing pattern is correct. Hence $\Omega$ is a Moebius band (rather than a cylinder).

Finally, we triangulate each of our 4 rectangles by adding in a diagonal, as in Figure 2 (right). The long sides of the thin triangles all have length at least $\mu + \epsilon^2$ and the short sides have length less than $2\epsilon^3$.

**The Domain:** Call a clean triangle *wide* if it is isosceles and its distinguished side is longer than $\mu$. The distinguished side of a wide clean triangle is longer than the other two sides. Hence there is a clean triangle $T'_0$ whose distinguished side has length in $(\mu+\epsilon^2, \mu+2\epsilon^2)$ and whose other sides have length less than $\mu+\epsilon^2$. We normalize so that the distinguished edge of $T'_0$ lies in the top boundary component of $\Sigma$. Let $T'_\pm$ be clean and congruent triangles on either side of $T'_0$ such that the gap between $T'_0$ and $T'_\pm$ is a parallelogram $R'_\pm$ having short side length $2\epsilon^2$. Let $S'_\pm$ denote the parallelograms having short side length $\epsilon^2$ that lie on either side of $T'_\pm$. Figure 3 shows the union of these three triangles and 4 parallelograms.

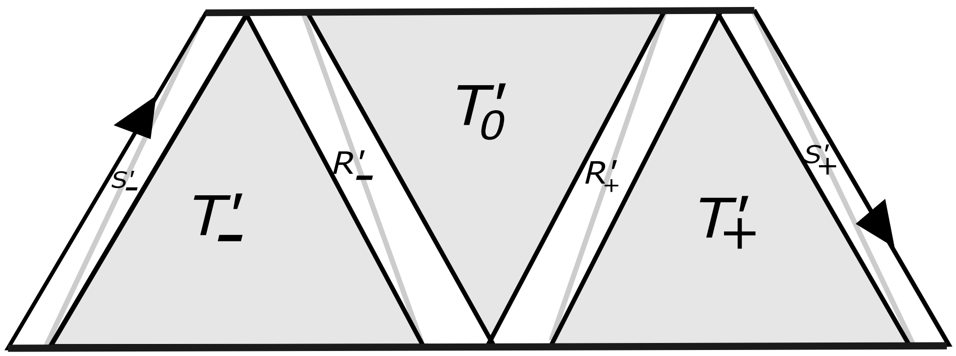


**Figure 3:** The domain $\Omega' = \Sigma/G_\beta$ and its triangulation.

Note that $3\mu/2 = \sqrt{3}$. Let $G_\beta$ be the glide reflection that identifies the left side of $S'_-$ with the right side of $S'_+$. The midline of $\Sigma$ intersects each of the three triangles in Figure 3 in a segment of length at most $\mu/2+\epsilon^2$ and two of the parallelograms in a segment of length $\epsilon^2$, and two of the parallelograms in a segment of length $2\epsilon^2$. Hence $\beta < \sqrt{3}+\epsilon$. The union of polygons in Figure 3 is a fundamental domain for the action of $G_\beta$ on $\Sigma$. The quotient $\Omega' = \Sigma/G_\beta$ is a flat Moebius band.

**The Map:** As in Figure 3, we triangulate the parallelograms in $\Omega'$ so as to always use the short diagonals. Each of these diagonals is shorter than the long sides of the parallelograms. Hence, each of these diagonals has length less than $\mu+\epsilon^2$. The triangulations of $\Omega'$ and $\Omega$ are combinatorially identical and hence determine an injective and continuous piecewise affine map $\overline{f}$ from $\Omega'$ to $\Omega$. Let's check the squeeze condition.

- On each $T'_k$, for $k \in \{0,+,-\}$, the crosscutting edges have length less than $\mu+\epsilon^2$ and the distinguished edge has length greater than $\mu+\epsilon^2$. The map $\overline{f}$ carries all these edges to edges of length exactly $\mu+\epsilon^2$. Hence $f$ has the squeeze property on $T'_k$.

- On each thin triangle, the crosscutting edges (again) have length less than $\mu+\epsilon^2$ and the distinguished edges have length at least $\epsilon^2$. The map $\overline{f}$ carries the crosscutting edges to edges of length at least $\mu+\epsilon^2$ and the distinguished edges to edges of length at most $2\epsilon^3 < \epsilon^2$. Hence $\overline{f}$ has the squeeze property on all thin triangles.

Therefore, the squeeze condition holds on every triangle. Finally, we let $f: \Sigma \to \Omega$ be the equivalent lift of $\overline{f}$. More precisely, $f = \overline{f} \circ \pi$, where $\pi: \Sigma \to \Omega'$ is the quotient map. This map has all the required properties. Hence $\beta < \sqrt{3}+\epsilon$ is realized. This proves Statement 2.